



SHAKTI

Bitstream Generation & FPGA Configuration

SHAKTI Group | CSE Dept | PS-CDISHA - RISE LAB

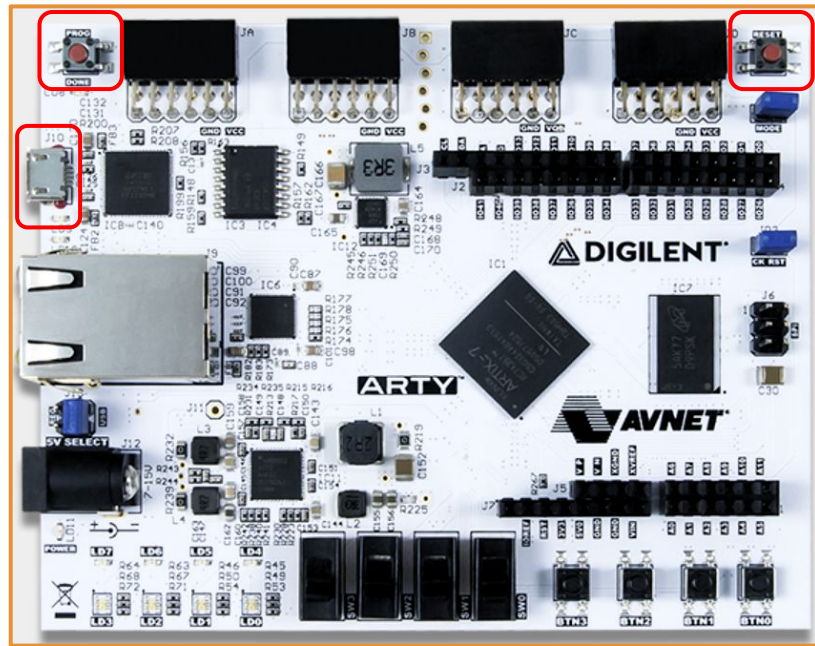
IIT Madras

Webinar Schedule

1.	Introduction
2.	Recap of Tool Setup
3.	Flow of Bitstream/MCS Generation and Board Configuration
4.	TCL-based Bitstream & MCS Generation and Configuration
5.	FPGA Configuration: Pre-built Bitstream and MCS Files
6.	FPGA Configuration: Bitstream & MCS via Vivado GUI
7.	FPGA Configuration: Status Check

Introduction

- Reconfigurable IC's
- Bitstream File
 - Volatile, Reset, Program
 - Approx ~6 seconds
- MCS File (Memory Configuration Storage)
 - Non-Volatile 128Mb QSPI Flash Memory
 - Approx 4 - 5 Minutes
- Generating Bitstream & MCS
- Configuring Bitstream & MCS
- Ensuring the tool setup is complete is essential before generating these binary files.



ARTY_A7_100T FPGA Board

<https://digilent.com/reference/programmable-logic/arty-a7/start>

FPGA Configuration: Bitstream & MCS

→ Bitstream

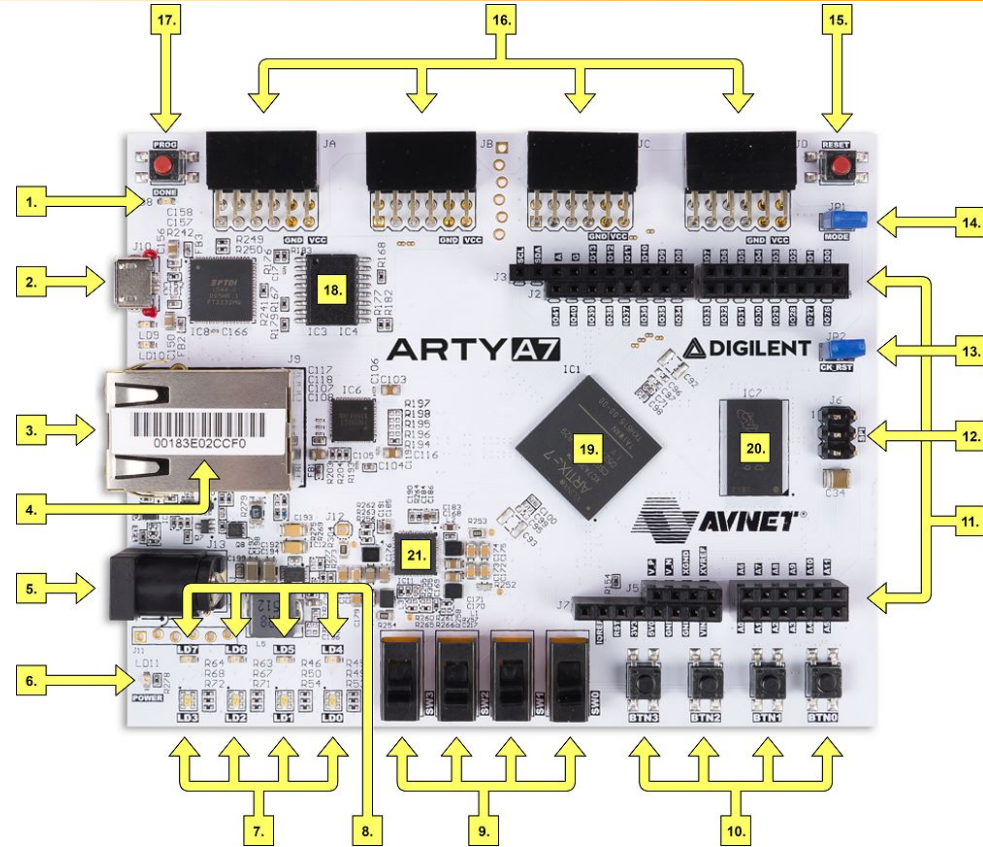
- Volatile, Reset, Program
- Approx ~6 seconds

→ MCS

- Non-Volatile 128Mb Serial Flash Memory
- X4 SPI
- Approx 4 - 5 Minutes

→ Configuring ARTY A7 100T FPGA Board either in two ways:

- JTAG Configuration (J10 USB - JTAG)
- Quad-SPI Configuration

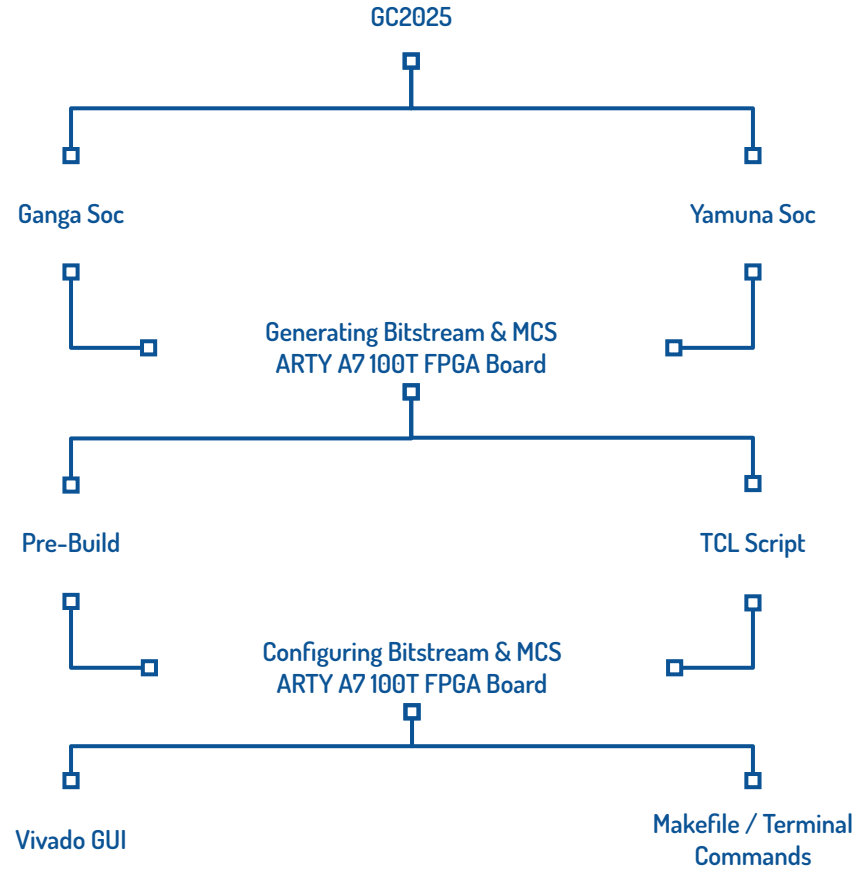


reference:programmable-logic:arty-a7:arty-a7-callout.png

Recap of Tool Setup

- Make sure all the hardware and software dependencies should installed (Tool Setup Manual) before bitstream and mcs generation
- Clone the gc2025 repo from the shakti gitlab by using the following link
 - <https://gitlab.com/shaktiproject/gc2025.git>

Flow of Bitstream/MCS Generation and Board Configuration



TCL-based Bitstream & MCS Generation

- SHAKTI uses Makefile and TCL scripts to generate bitstream and MCS files for target FPGA boards.
- Follow the steps to generate bitstream and mcs for the target fpga board
 - Open the Ubuntu terminal and navigate to the gc2025/hw directory.
 - Refer to the [README.md](#) file in the hw folder or the document “Bitstream/MCS Generation and FPGA Target Configuration.doc” for exact commands to build bitstream and MCS files.
- For Ganga SoC
 - make build BOARD=arty_a7_ganga
- For Yamuna SoC
 - make build BOARD=arty_a7_yamuna XLEN_build=32

TCL-based Bitstream & MCS Generation

```
## puts "DEBUG: ip_project_dir: $ip_project_dir"
DEBUG: ip_project_dir: ./fpga_project/manage_ip
## puts "DEBUG: core_project_dir: $core_project_dir"
DEBUG: core_project_dir: ./fpga_project/c-class
# set jobs [lindex $argv 0]
# open_project $core_project_dir/$core_project.xpr
Scanning sources...
Finished scanning sources
INFO: [IP_Flow 19-234] Refreshing IP repositories
INFO: [IP_Flow 19-1704] No user IP repositories specified
INFO: [IP_Flow 19-2313] Loaded Vivado IP repository '/scratch/vivado_18/Vivado/2018.3/data/ip'.
# set top_module [get_property TOP [get_filesets sources_1]]
# set bit $core_project_dir/$core_project.runs/core_impl_1/$top_module.bit
# set mcs $core_project_dir/$top_module.mcs
# set bitload "up 0x00000000 $bit"
# puts "BITSTREAM: $bit MCS: $mcs"
BITSTREAM: ./fpga_project/c-class/c-class.runs/core_impl_1/fpga_top.bit MCS: ./fpga_project/c-class/fpga_top.mcs
# write_cfgmem -format mcs -size 16 -interface SPIx1 -loadbit $bitload -force -file $mcs;
Command: write_cfgmem -format mcs -size 16 -interface SPIx1 -loadbit {up 0x00000000 ./fpga_project/c-class/c-class.runs/core_impl_1/fpga_top.bit} -force -file ./fpga_project/c-class/fpga_top
.mcs
Creating config memory files...
Creating bitstream load up from address 0x00000000
Loading bitfile ./fpga_project/c-class/c-class.runs/core_impl_1/fpga_top.bit
Writing file ./fpga_project/c-class/fpga_top.mcs
Writing log file ./fpga_project/c-class/fpga_top.prm
=====
Configuration Memory information
=====
File Format      MCS
Interface        SPIx1
Size             16M
Start Address    0x00000000
End Address      0x00FFFFFF

Addr1      Addr2      Date      File(s)
0x00000000  0x003A607B  Oct  7 01:09:08 2025  ./fpga_project/c-class/c-class.runs/core_impl_1/fpga_top.bit
0 Infos, 0 Warnings, 0 Critical Warnings and 0 Errors encountered.
write_cfgmem completed successfully
# puts "MCS Generated !"
MCS Generated !
# exit
```

➔ After successfully generating the bitstream and MCS files, the terminal displays a success message confirming the creation of both files.

TCL-based Bitstream Configuration

→ Configuring Bitstream for Ganga SoC

- make program_bitstream BOARD=arty_a7_ganga

→ Configuring Bitstream for Yamuna SoC

- make program_bitstream BOARD=arty_a7_yamuna

```
DEBUG: ip_project_dir: ./fpga_project/manage_ip
## puts "DEBUG: core_project_dir: $core_project_dir"
DEBUG: core_project_dir: ./fpga_project/c-class
# set jobs [lindex $argv 0]
# open_project $core_project_dir/$core_project.xpr
Scanning sources...
Finished scanning sources
INFO: [IP_Flow 19-234] Refreshing IP repositories
INFO: [IP_Flow 19-1704] No user IP repositories specified
INFO: [IP_Flow 19-2313] Loaded Vivado IP repository '/scratch/vivado_18/Vivado/2018.3/data/ip'.
# set top_module [get_property TOP [get_filesets sources_1]]
# set bit $core_project_dir/$core_project.runs/core_impl_1/$top_module.bit
# set device [lindex $argv 1]
# puts "BITSTREAM: $bit"
BITSTREAM: ./fpga_project/c-class/c-class.runs/core_impl_1/fpga_top.bit
# open_hw
# connect_hw_server
INFO: [Labtools 27-2285] Connecting to hw_server url TCP:localhost:3121
INFO: [Labtools 27-2222] Launching hw_server...
INFO: [Labtools 27-2221] Launch Output:

***** Xilinx hw_server v2018.3
**** Build date : Dec 6 2018-23:53:53
** Copyright 1986-2018 Xilinx, Inc. All Rights Reserved.

# open_hw_target
INFO: [Labtoolstcl 44-466] Opening hw_target localhost:3121/xilinx_tcf/Digilent/210319AD2CE0A
# set_property PROGRAM.FILE $bit [lindex [get_hw_devices] 0]
# program_hw_devices [current_hw_device]
INFO: [Labtools 27-3164] End of startup status: HIGH
# disconnect_hw_server
# puts "INFO: FPGA has been programmed with the given bitstream"
INFO: FPGA has been programmed with the given bitstream
# exit
INFO: [Common 17-206] Exiting Vivado at Wed Oct 8 19:49:00 2025...
echo "Bitstream programmed to FPGA."
Bitstream programmed to FPGA.
echo "Run \"sudo openocd -f shakti-arty.cfg\" to start GDB server at localhost:3333"
Run "sudo openocd -f shakti-arty.cfg" to start GDB server at localhost:3333
```

TCL-based MCS Configuration

→ Configuring MCS for Ganga SoC

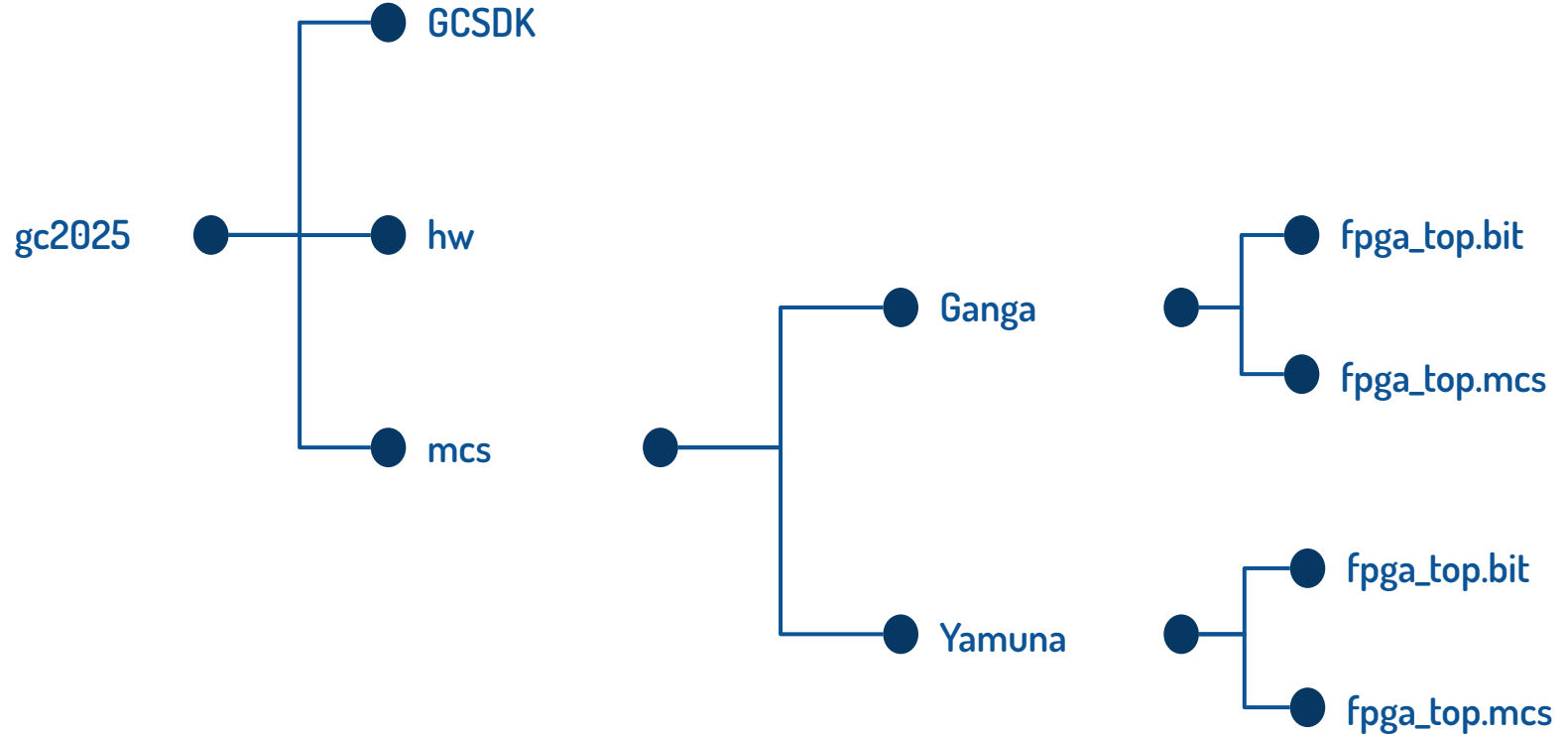
- make program_mcs BOARD=arty_a7_ganga

→ Configuring MCS for Yamuna SoC

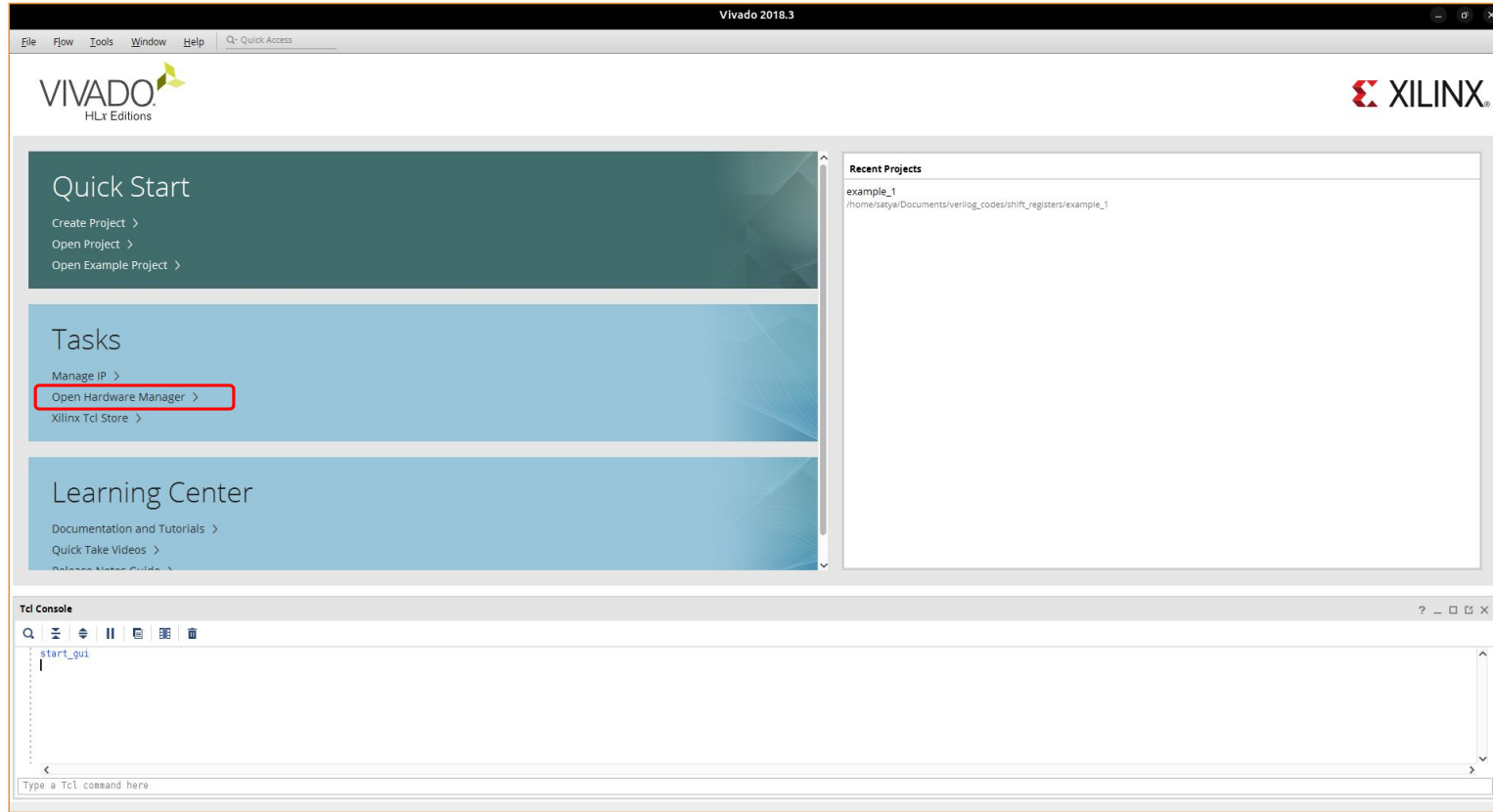
- make program_mcs BOARD=arty_a7_yamuna

```
INFO: [Xicom 50-104] The verify operation will only be performed on the address range specified by the Memory Configuration File (MCS).
Mfg ID : 1   Memory Type : 20   Memory Capacity : 18   Device ID 1 : 0   Device ID 2 : 0
Performing Erase Operation...
Erase Operation successful.
Performing Program and Verify Operations...
Program/Verify Operation successful.
INFO: [Labtoolstcl 44-377] Flash programming completed successfully
program_hw_cfgmem: Time (s): cpu = 00:00:00.21 ; elapsed = 00:01:45 . Memory (MB): peak = 2180.531 ; gain = 3.656 ; free physical = 24827 ; free virtual = 42942
# endgroup
# disconnect_hw_server localhost:3121;
# puts "INFO: FPGA CFG MEM has been programmed with the given bitstream"
INFO: FPGA CFG MEM has been programmed with the given bitstream
# puts "Please Disconnect FPGA from USB Port to use"
Please Disconnect FPGA from USB Port to use
# exit
INFO: [Common 17-206] Exiting Vivado at Wed Oct  8 19:55:22 2025...
echo "Please Disconnect and reconnect Your Arty Board from your PC"
Please Disconnect and reconnect Your Arty Board from your PC
echo "After programming reset the device once and run \"sudo openocd \
-f shakti-arty.cfg\" to start a gdb server at localhost:3333 "
After programming reset the device once and run "sudo openocd -f shakti-arty.cfg" to start a gdb server at localhost:3333
```

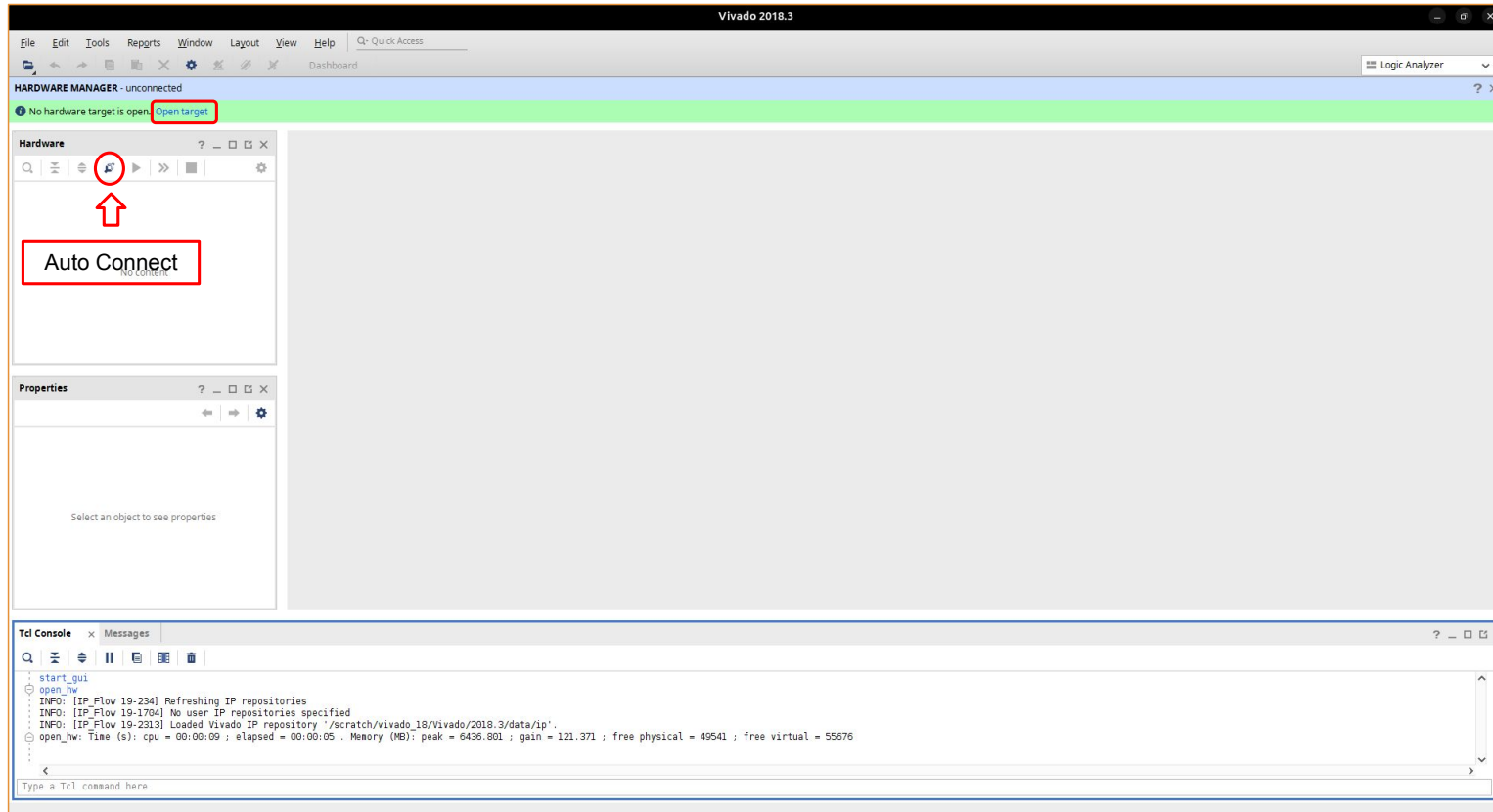
FPGA Configuration: Pre-built Bitstream and MCS Files



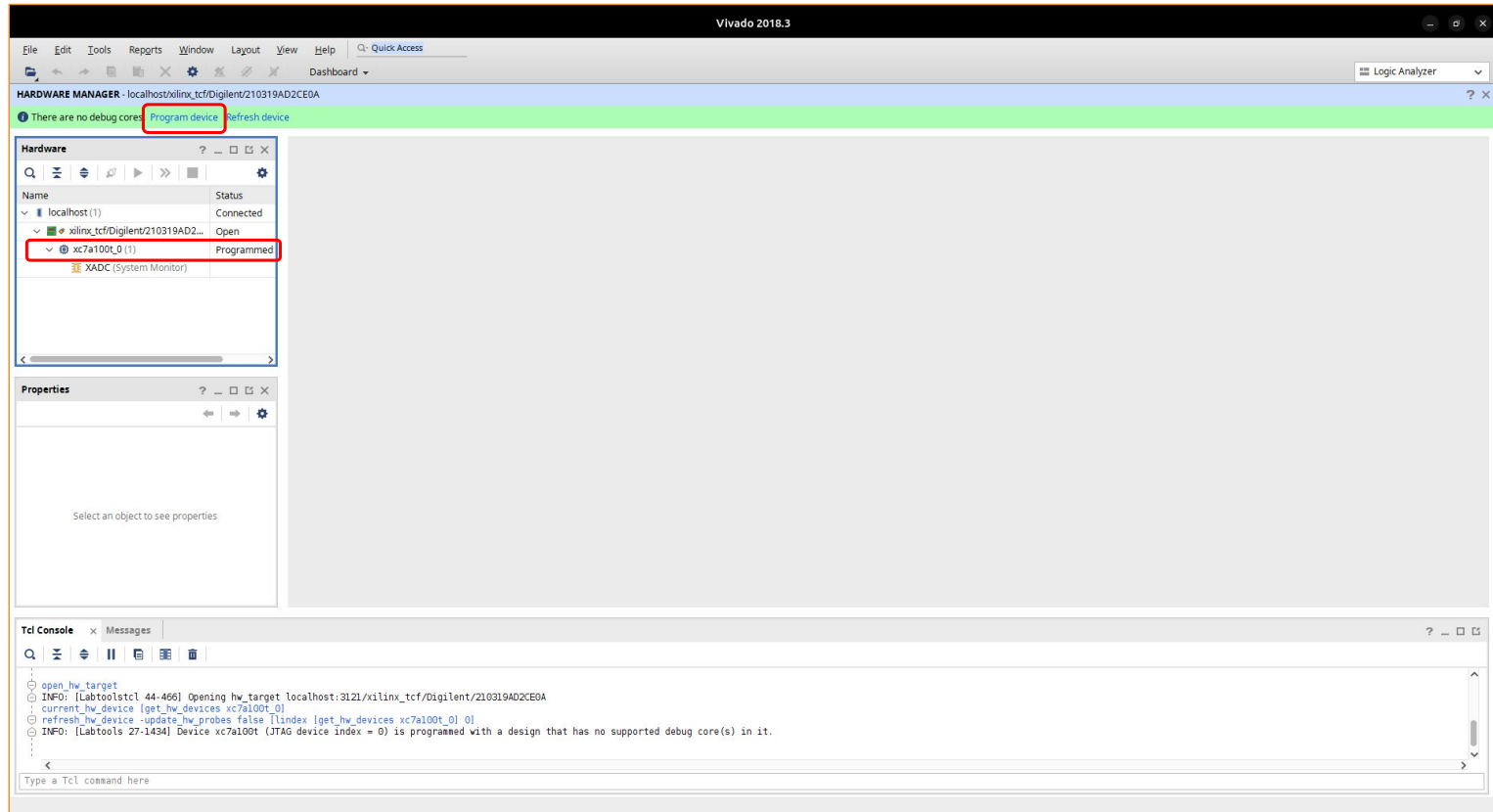
FPGA Configuration: Loading Bitstream via Vivado GUI



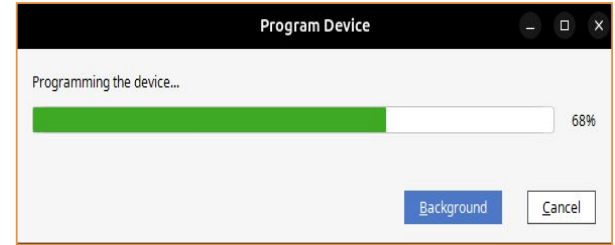
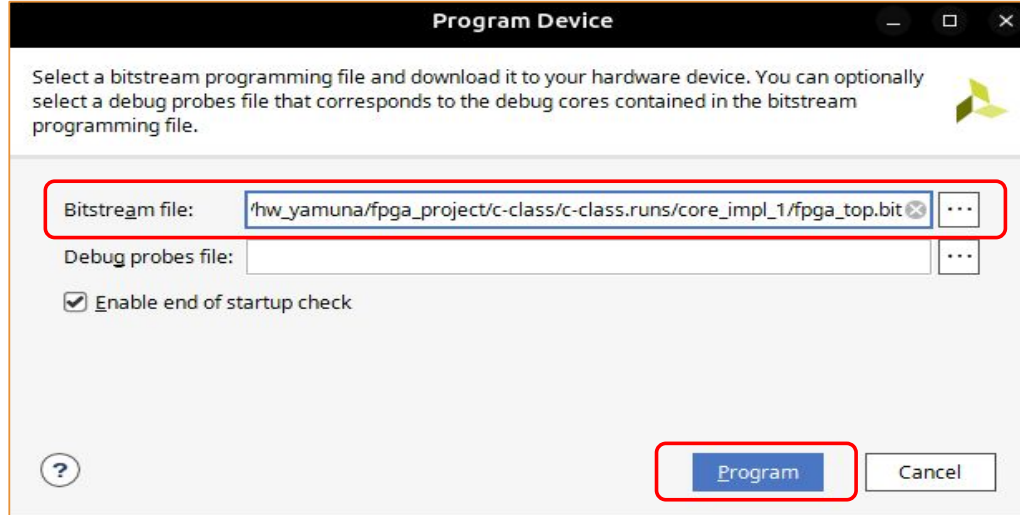
FPGA Configuration: Loading Bitstream via Vivado GUI



FPGA Configuration: Loading Bitstream via Vivado GUI



FPGA Configuration: Loading Bitstream via Vivado GUI



FPGA Configuration: Loading Bitstream via Vivado GUI

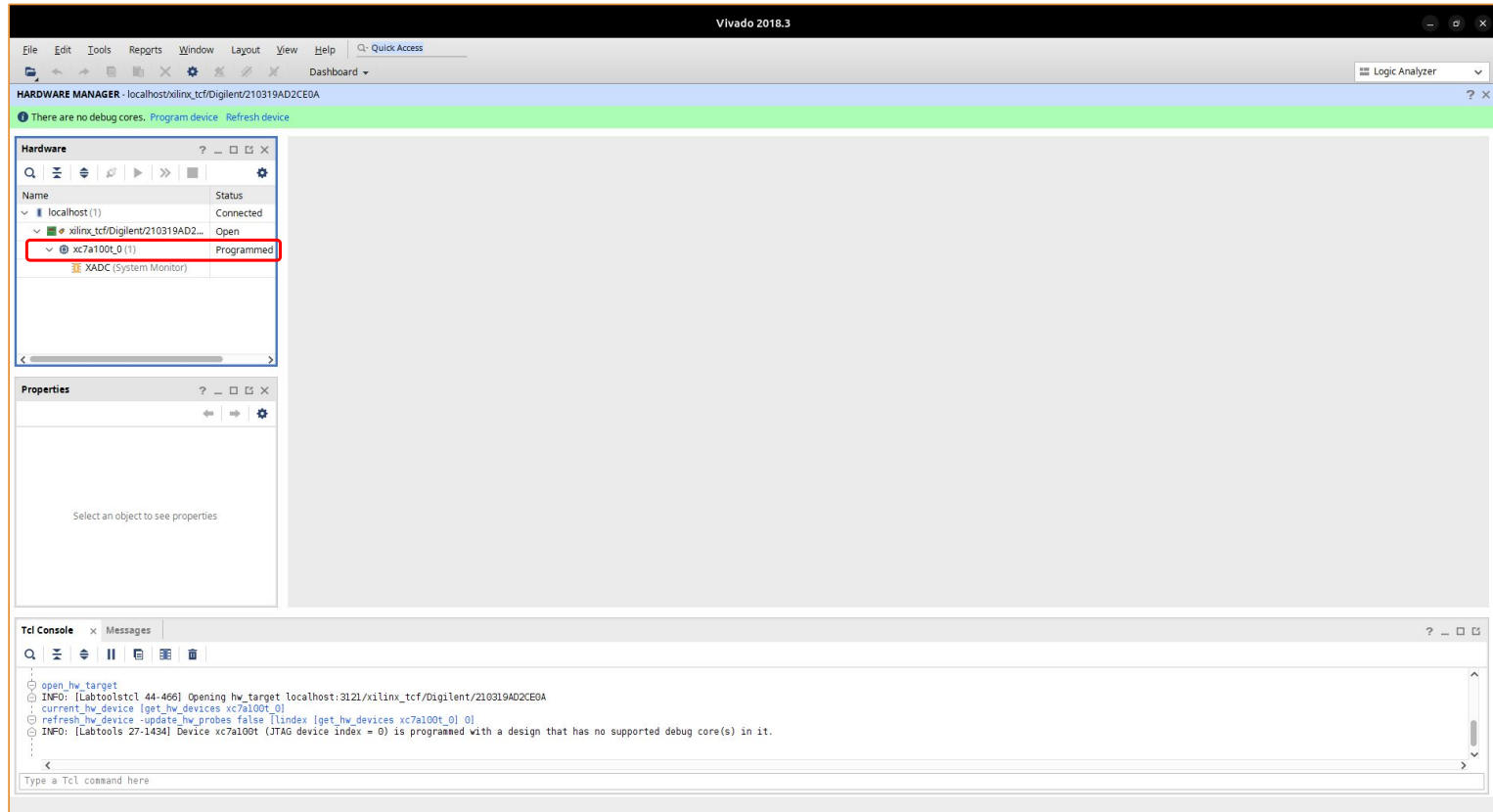
```
Tcl Console x Messages
[Icons]

start_gui
open_hw
INFO: [IP_Flow 19-234] Refreshing IP repositories
INFO: [IP_Flow 19-1704] No user IP repositories specified
INFO: [IP_Flow 19-2313] Loaded Vivado IP repository '/scratch/vivado_18/Vivado/2018.3/data/ip'.
connect_hw_server
INFO: [Labtools 27-2285] Connecting to hw_server url TCP:localhost:3121
INFO: [Labtools 27-2222] Launching hw_server...
INFO: [Labtools 27-2221] Launch Output:

***** Xilinx hw_server v2018.3
**** Build date : Dec 6 2018-23:53:53
** Copyright 1986-2018 Xilinx, Inc. All Rights Reserved.

open_hw_target
INFO: [Labtoolstcl 44-466] Opening hw_target localhost:3121/xilinx_tcf/Digilent/210319AD2CE0A
current_hw_device [get_hw_devices xc7a100t_0]
refresh_hw_device -update_hw_probes false [lindex [get_hw_devices xc7a100t_0] 0]
INFO: [Labtools 27-1434] Device xc7a100t (JTAG device index = 0) is programmed with a design that has no supported debug core(s) in it.
refresh_hw_device [lindex [get_hw_devices xc7a100t_0] 0]
INFO: [Labtools 27-1434] Device xc7a100t (JTAG device index = 0) is programmed with a design that has no supported debug core(s) in it.
set_property PROBES.FILE {} [get_hw_devices xc7a100t_0]
set_property FULL_PROBES.FILE {} [get_hw_devices xc7a100t_0]
set_property PROGRAM.FILE {/p/gc2025/mcs/Ganga/ArtyA7100T_26082025.bit} [get_hw_devices xc7a100t_0]
program_hw_devices [get_hw_devices xc7a100t_0]
INFO: [Labtools 27-3164] End of startup status: HIGH
refresh_hw_device [lindex [get_hw_devices xc7a100t_0] 0]
INFO: [Labtools 27-1434] Device xc7a100t (JTAG device index = 0) is programmed with a design that has no supported debug core(s) in it.
```


FPGA Configuration: Loading MCS via Vivado GUI



FPGA Configuration: Loading MCS via Vivado GUI

Add Configuration Memory Device

Choose a configuration memory part.

Device: xc7a100t_0

Filter

Manufacturer: Spansion
Density (Mb): 128
Type: spi
Width: x1_x2_x4
[Reset All Filters](#)

Select Configuration Memory Part

Search: Q-

Name	Part	Manufact...	Alias	Family	Type	Density (...)	Width
s25fl128l-spi-x1_x2_x4	s25fl128l	Spansion		s25flxxxl	spi	128	x1_x2_x4
s25fl128sxxxxxx0-spi-x1_x2_x4	s25fl128sxxxxxx0	Spansion	s25fl127s-spi-x1_x2_x4	s25flxxxs	spi	128	x1_x2_x4
s25fl128sxxxxxx1-spi-x1_x2_x4	s25fl128sxxxxxx1	Spansion		s25flxxxs	spi	128	x1_x2_x4

?

OK Cancel

FPGA Configuration: Loading MCS via Vivado GUI

The screenshot displays the Vivado 2018.3 interface. The **HARDWARE MANAGER** tab is active, showing a tree view of hardware components. The component **s25fl256sxxxxxx0-spi-x1_x2_x4** is selected and highlighted with a red box. Below the tree view, the **Configuration Memory Device Properties** window is open, showing the same component name and memory part. A dialog box titled **Add Configuration Memory Device Completed** is displayed in the center, asking: "Do you want to program the configuration memory device now?". The **OK** button is highlighted with a red box. At the bottom, the **Tcl Console** shows a list of commands and a message: "INFO: [Labtools 27-1434] Device xc7a100t (JTAG device index = 0) is programmed with a design that has no supported debug core(s) in it." The command **Configuration Memory Device: s25fl256sxxxxxx0-spi-x1_x2_x4** is entered in the console input field and highlighted with a red box.

Vivado 2018.3

File Edit Tools Repgrts Window Layout View Help Quick Access

HARDWARE MANAGER - localhost/xilinx_tcf/Digilent/210319AD2CE0A

There are no debug cores. Program device Refresh device

Hardware

localhost (1) Status Connected

xilinx_tcf/Digilent/210319AD2CE0A Open

xc7a100t_0 (2) Programmed

XADC (System Monitor)

s25fl256sxxxxxx0-spi-x1_x2_x4

Configuration Memory Device Properties

s25fl256sxxxxxx0-spi-x1_x2_x4

Name: s25fl256sxxxxxx0-spi-x1_x2_x4

Memory Part: s25fl256sxxxxxx0-spi-x1_x2_x4

Memory type: spi

Memory density: 256

Programming file:

Add Configuration Memory Device Completed

Do you want to program the configuration memory device now?

☐ Don't show this dialog again

OK Cancel

Tcl Console

set_property PROGRAM.ERASE 1 [get_property PROGRAM.HW_CFGMEM [lindex [get_hw_devices xc7a100t_0] 0]]

set_property PROGRAM.CFG_PROGRAM 1 [get_property PROGRAM.HW_CFGMEM [lindex [get_hw_devices xc7a100t_0] 0]]

set_property PROGRAM.VERIFY 1 [get_property PROGRAM.HW_CFGMEM [lindex [get_hw_devices xc7a100t_0] 0]]

set_property PROGRAM.CHECKSUM 0 [get_property PROGRAM.HW_CFGMEM [lindex [get_hw_devices xc7a100t_0] 0]]

refresh_hw_device [lindex [get_hw_devices xc7a100t_0] 0]

INFO: [Labtools 27-1434] Device xc7a100t (JTAG device index = 0) is programmed with a design that has no supported debug core(s) in it.

Type a Tcl command here

Configuration Memory Device: s25fl256sxxxxxx0-spi-x1_x2_x4

FPGA Configuration: Loading MCS via Vivado GUI

Program Configuration Memory Device

Select a configuration file and set programming options.

Memory Device: ...

Configuration file: ...

PRM file: ...

State of non-config mem I/O pins:

Program Operations

Address Range:

☒ Erase

☐ Blank Check

☒ Program

☒ Verify

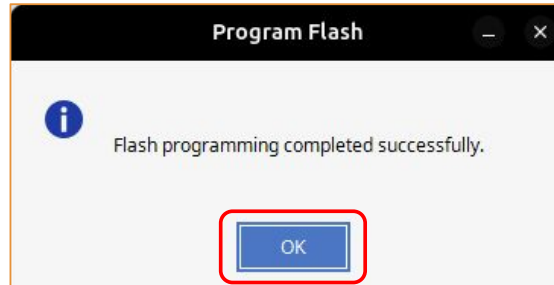
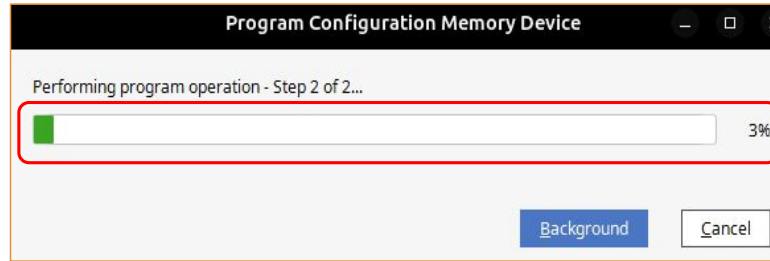
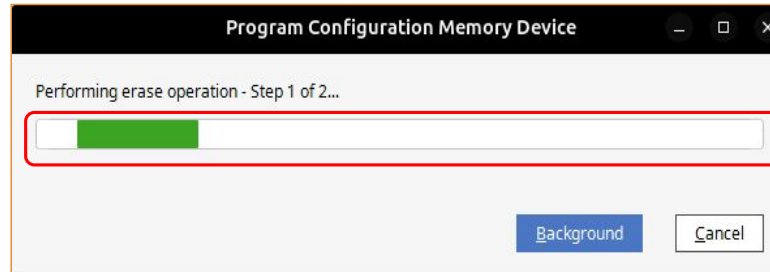
☐ Verify Checksum

SVF Options

☐ Create SVF Only (no program operations)

SVF File: ...

FPGA Configuration: Loading MCS via Vivado GUI



FPGA Configuration: Status Check



Queries ?



Thank You !

